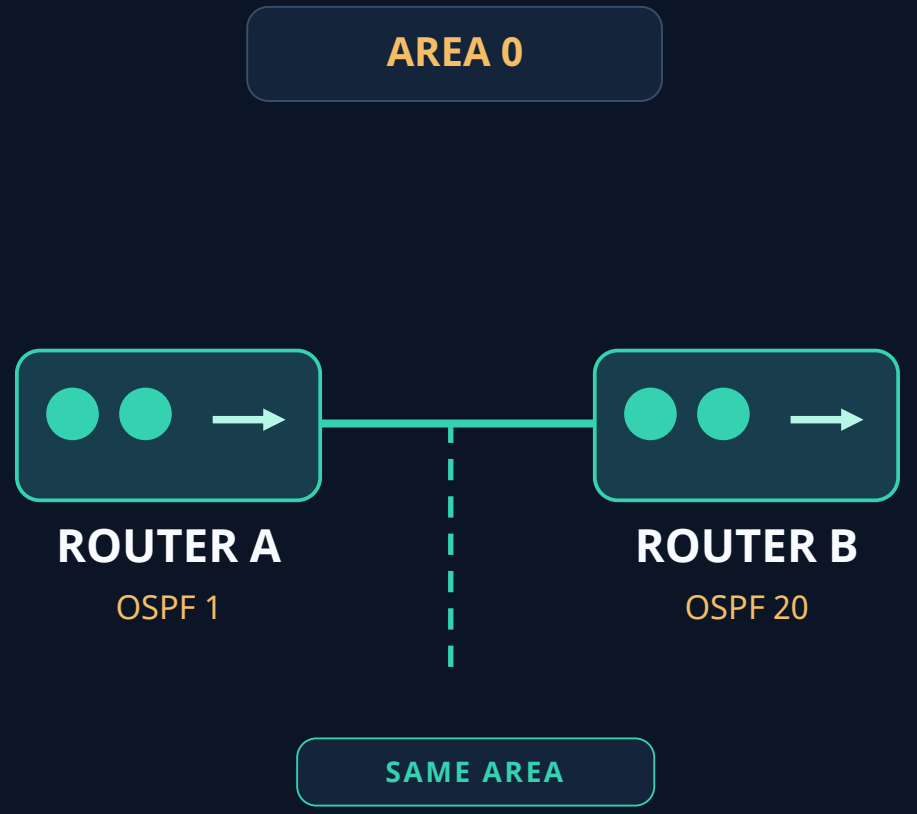
NETWORKING • CISCO IOS

OSPF Commands Explained

When do the process ID, network, wildcard mask, and area numbers change?



enable → config terminal → router ospf 1

The five commands

Read the whole configuration first, then break down each value.

CISCO IOS CONFIGURATION

```
enable
config terminal
router ospf 1
network 0.0.0.0 255.255.255.255 area 0
exit
```

1. Enter admin mode
2. Enter configuration
3. Start OSPF process
4. Select interfaces & area
5. Leave OSPF mode

The network line is the one we change most often.

Move through Cisco IOS modes

The prompt shows your current mode.

USER EXEC

Router>

Type: enable

Router#

→

PRIVILEGED EXEC

Router#

Type: configure terminal

Router(config)#

→

GLOBAL CONFIG

Router(config)#

Type: router ospf 1

Router(config-router)#

exit from Router(config-router)# returns to Router(config)#

What does router ospf 1 mean?

The number is a local process ID—not a shared password or area number.

```
router ospf 1
```

You can also use 10, 20 or 100.

- It identifies an OSPF process on this router.
- Neighboring routers may use different IDs.
- Change it if managing a different local process.



LOCAL ID ≠ NEIGHBOR ID

Understand the network statement

This statement controls which IPv4 interfaces join OSPF and their area.

```
network 0.0.0.0 255.255.255.255 area 0
```

NETWORK ADDRESS

0.0.0.0

Start of the address-matching pattern

WILDCARD MASK

255.255.255.255

All address bits are “don’t care”

AREA ID

0

Put matched interfaces into Area 0

Result: OSPF runs on all IPv4-addressed interfaces in this process.

Wildcard masks: 0 vs 1

A wildcard mask is not a subnet mask. It is used to match addresses.

WILDCARD BIT 0

MUST MATCH

0.0.0.0 wildcard → match exactly

WILDCARD BIT 1

ANY VALUE

255.255.255.255 → match all

Example **192.168.1.0** + **0.0.0.255** → **match 192.168.1.***

Quick rule: wildcard = 255.255.255.255 minus subnet mask (octet by octet).

When should the values change?

Choose a network match and area based on the design—not by memorizing one line.

1 • All IPv4 interfaces

network 0.0.0.0
255.255.255.255 area 0

Broad matching; simple labs

2 • One /24 range

network 192.168.1.0
0.0.0.255 area 0

Interfaces whose IPs are in this /24

3 • One exact IP

network 192.168.1.1
0.0.0.0 area 0

Only the interface with this address

4 • A different area

network 10.0.0.0
0.0.0.255 area 1

Place matched interfaces in Area 1

Subnet mask → wildcard mask

For typical contiguous subnet masks, invert every octet.

SUBNET MASK	WILDCARD MASK	USE
255.255.255.255 (/32)	**0.0.0.0**	One exact address
255.255.255.0 (/24)	**0.0.0.255**	One /24 range
255.255.0.0 (/16)	**0.0.255.255**	One /16 range
255.0.0.0 (/8)	**0.255.255.255**	One /8 range
0.0.0.0 (/0)	**255.255.255.255**	All IPv4 addresses

Remember: 0 = check this bit | 1 = ignore this bit

Two-router example: different process IDs

The OSPF interface area must match; the local process IDs do not.



ROUTER 1

```
router ospf 1
network 192.168.1.0 0.0.0.255 area 0
```

ROUTER 2

```
router ospf 20
network 192.168.1.0 0.0.0.255 area 0
```

They can form an OSPF adjacency when their other OSPF settings are compatible.

Area IDs: when can they differ?

On a shared OSPF link, both endpoints must agree on the area.

MATCHING AREAS

R1: area 0

R2: area 0

Area setting is compatible

MISMATCHED AREAS

R1: area 0

R2: area 1

OSPF adjacency will not form

Area 0 is the OSPF backbone. Area 1, Area 2, etc. are used in multi-area designs.

Verify the configuration

After configuring OSPF, check its neighbors and learned routes.

show ip ospf neighbor	Are OSPF neighbors forming?
show ip protocols	Which process and networks are configured?
show ip route ospf	Which routes were learned through OSPF?
show ip ospf interface brief	Which interfaces participate in OSPF?

Best practice: match only the needed interfaces; use passive-interface where appropriate.

Quick knowledge check

Ask students to predict the correct OSPF command.

1 **Enable OSPF on every IPv4 interface in Area 0.**

network 0.0.0.0 255.255.255.255 area 0

2 **Enable only the interface with IP 10.1.1.1 in Area 0.**

network 10.1.1.1 0.0.0.0 area 0

3 **Match 172.16.5.0/24 and place it in Area 2.**

network 172.16.5.0 0.0.0.255 area 2

Key takeaways

The three numbers have three different jobs.

01 PROCESS ID — **router ospf 1**

Local identifier; neighboring routers can use different IDs.

02 NETWORK + WILDCARD — **network 192.168.1.0 0.0.0.255**

Choose which interface IP addresses OSPF matches.

03 AREA ID — **area 0**

Choose the OSPF area; shared-link neighbors must match.

Configure → Verify → Troubleshoot